FINAL YEAR PROJECT · UI/UX DESIGN PROPOSAL

PARALLAX

UI/UX proposal for a 70 m² Signal & Computer Vision Lab

Local interactive edition, secondary research and conceptual physical design



Conceptual P4-D room render. Authored project visualization, not a photograph of an installed laboratory.

A current assessor entry links the working app, Figma designs, physical model, source evidence, presentation and static demonstration. This report distinguishes authored examples from measurements and institution-owned services.

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AT A GLANCE

The proposal in nine numbers

One 70 m² room, designed as a physical space and a digital service together.

ROOM

70 m²

An assumed 10 × 7 m envelope, 3 m high, divided into seven zones.

BOOKABLE RESOURCES

13

Stations, the capture area, camera, lights, tripod, target and signal bench.

REVIEWED SOURCES

14

Secondary research only. Zero local participants.

REQUIREMENTS

20

Each traced to a problem, a wireframe and an implementation boundary.

WIREFRAMES

15

Annotated boards from preparation to technician recovery.

DESIGN SYSTEM

40

Component families, with 87 variants in the Figma library.

SMART FEATURES

9

Five working, two simulated, one partly built, one concept.

SYSTEM SCENARIOS

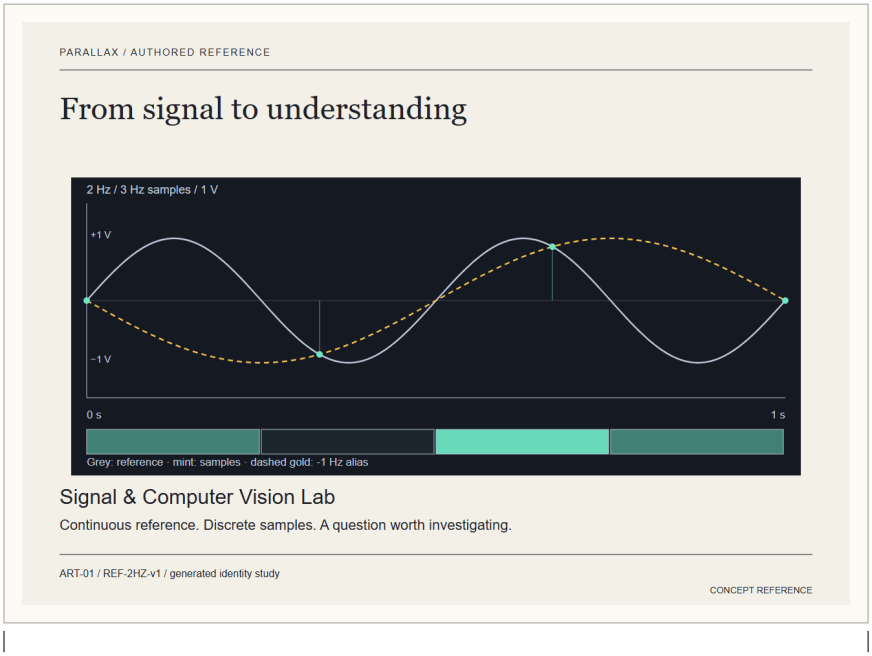
14

Conflicts, retries, faults and fallbacks, each reconciled.

AUTOMATED TESTS

155

139 application tests and 16 assistant-service tests pass.



The lab's identity study: a 2 Hz reference, and the four samples a 3 Hz rate keeps of it. Generated artwork, not a measurement.

SECTION 01

1. Introduction

PARALLAX brings the physical laboratory and its digital workflow into one proposal. Students prepare an experiment, reserve its complete kit, identify the station, retain evidence, explain a comparison and close the session. Faculty review readiness and learning records. Technicians follow equipment impact, verification and handover. A separate public view communicates useful status without exposing private records.

The assignment supplies a 70 m² area. P4-D uses an assumed 10 × 7 m inner envelope with a 3 m height. The proposal combines computing, controlled image capture, a signal bench, collaboration, faculty support, storage and a clear circulation spine. These dimensions establish a reproducible design model. No site survey or occupied-room trial has established actual capacity, daylight, ventilation, services or access.

The delivered system is a runnable local React prototype supported by preserved Figma designs, fifteen wireframes, Blender source, generated scientific demonstrations and documented recovery paths. The smart dashboard and energy ledger use declared example records. The assistant performs real inference through local Ollama but reads only reviewed public project guides. The intended contribution is a coherent, inspectable proposal with explicit source and authority boundaries. Learning improvement, actual energy savings and institutional operating approval remain unmeasured.



FIG. 01 Conceptual P4-D room render. Authored project visualization, not a photograph of an installed laboratory.

Workspace evidence: PARALLAX_EXCELLENCE_BLUEPRINT.md; spatial/lab_geometry.json; planning/EXCELLENCE_EXECUTION_PHASES.md

SECTION 02

2. Problem Analysis

The design problem spans scientific reasoning, shared resources and physical access. A student can hold a station reservation while a required accessory is missing. A displayed curve can look convincing while its sampling conditions remain ambiguous. A saved draft can appear complete before a submission receipt exists. An available asset can still be unsuitable for a task, and a recent-looking dashboard value can have a disconnected source.

These are prospective failure consequences and authored scenarios. The project does not establish their frequency in the intended cohort. The research traceability records twenty requirements, twelve assumptions and eight design decisions. Each requirement connects a problem, source rationale, wireframe and implementation boundary. Rejected shortcuts include a composite room-safety score, generic idle power-off, automatic physical setup transfer and an assistant that invents browser evidence.

The resulting design preserves distinctions that matter to a decision: availability versus occupancy, preparation versus training permission, recorded context versus current context, generated samples versus measured data, and draft versus confirmed receipt. Technical checks examine the implementation of these distinctions. A participant outcome or a real institution's policy cannot be inferred from passing domain tests.

PROBLEM	DESIGN RESPONSE	EVIDENCE
Incomplete shared kit	Reserve all dependencies for one interval	R01-R03; U07 tests
Evidence loses its context	Freeze source, method and settings at capture	R05-R07; U04/U05 tests
Recovery appears complete too early	Pending receipt and verification gates	R08-R13; migration tests
Unknown data looks authoritative	Source age, gap and manual fallback	R14-R20; smart-lab tests

Workspace evidence: research/U02_TRACEABILITY.csv; research/U02_HYPOTHESIS_REVIEW.md; research/U02_DESIGN_DECISIONS.md

SECTION 03

3. User Research

The agreed research method is secondary research and design-artifact analysis. Local participants: zero. Interviews, surveys, recruitment, participant usability sessions and field observation are excluded by the user's scope decision. Fourteen public sources were reviewed. This focused review is not a systematic literature review or a claim of saturation. The register distinguishes full text, abstract-only, indexed partial access and an unread correction.

Five education or provenance studies inform evidence comparison and record reconstruction. Two institutional operating pages inform training, booking and fault ownership. WCAG, ventilation guidance and LEAF inform digital access and sustainability questions. Two AI education studies inform bounded assistance. Two public discussion threads generate questions and counterexamples. External findings support a design rationale, while local behavior, pain frequency and learning effects remain unknown.

The six role lenses are first-run students, returning students, faculty, technicians, administrators and visitors. Personas and empathy maps are explicitly hypothetical. Their jobs concern preparation, reconstructing evidence, supporting a cohort, restoring equipment, reconciling resources and finding a public destination. Invented quotations and demographic preferences do not appear as findings. The journey follows discover, prepare, reserve, arrive, run, compare, reflect, submit and return, with named ownership at transitions.

The synthesis prioritizes six tensions: defending evidence, securing the whole kit, understanding recovery state, interpreting room context, retaining independent reasoning with AI and maintaining usable access alongside spatial richness. The ranking reflects possible consequence and assignment relevance. It does not use invented reach, frequency or benefit scores.

Workspace evidence: research/U02_SOURCE_REGISTER.md; research/U02_DESK_RESEARCH_METHOD.md; research/U02_RESEARCH_SYNTHESIS.md; research/PHASE_02_RESEARCH_AND_EXPERIENCE.md

SECTION 03 · RESEARCH FIGURES

User personas

Six role lenses frame who the lab serves. Each is a job to be done, with the local fact that would still have to be checked.

PA PRIMARY

First-run student

When a practical is unfamiliar, identify the method, complete kit and next action so I can begin with an understood question.

TRIGGER → OUTCOME
First preparation → explain the task and readiness requirements

STILL TO CHECK LOCALLY
Previous familiarity, how preparation happens, acceptable help and real stopping points

PB PRIMARY

Returning / advanced student

When conditions change, reconstruct and compare what I actually recorded so I can defend a bounded interpretation.

TRIGGER → OUTCOME
Variation, failed capture or later report → trace evidence without rewriting history

STILL TO CHECK LOCALLY
Current notebook habits, significance of parameter differences, preferred comparison depth

PC PRIMARY STAFF

Instructor

When preparing or teaching a cohort, distinguish readiness from learning so I can offer the right support.

TRIGGER → OUTCOME
Upcoming practical or confusing evidence → a support action, not a completion ranking

STILL TO CHECK LOCALLY
Assessment expectations, attendance ownership, what readiness signals actually mean

PD PRIMARY STAFF

Technician

When an asset fails or a kit returns incomplete, establish impact and ownership so the next user has a reliable status.

TRIGGER → OUTCOME
Incident or handover → restriction, repair verification and clear next action

STILL TO CHECK LOCALLY
Existing SOPs, notification responsibility, inventory accuracy and release authority

PE SECONDARY

Administrator

When reviewing demand and resources, reconcile definitions and scope so I can make a documented decision.

TRIGGER → OUTCOME
Planning review → qualified utilisation and resource evidence

STILL TO CHECK LOCALLY
Real reporting period, permitted data, baseline and denominators

PF TERTIARY

Visitor

When finding the lab or an activity, obtain useful location information without seeing private operational records.

TRIGGER → OUTCOME
Arrival and orientation → accessible destination and public explanation

STILL TO CHECK LOCALLY
Signage comprehension, public information policy and physical access

FIG. 02 Six role lenses, written as jobs to be done. Authored for testing; not validated biographies. Source: U02 research synthesis.

SECTION 03 · RESEARCH FIGURES

Empathy map

What each role is likely to say, think, do and feel at the moment the design has to help.

LIKELY SAYS – Which camera are we supposed to use? – Is this the right cable?	PA **First-run student** During the vision experiment	LIKELY THINKS – Everyone else seems to know what they're doing. – Did I save that?
LIKELY DOES – Checks the brief on a phone – Waits for the instructor – Retakes captures		LIKELY FEELS – Anxious at setup – Relieved after the first good capture – Uncertain about the report
PAINS – Incomplete kit – Unclear first step – Lost settings		GAINS – A clear checklist – Confidence before arriving – Evidence that writes the report's method

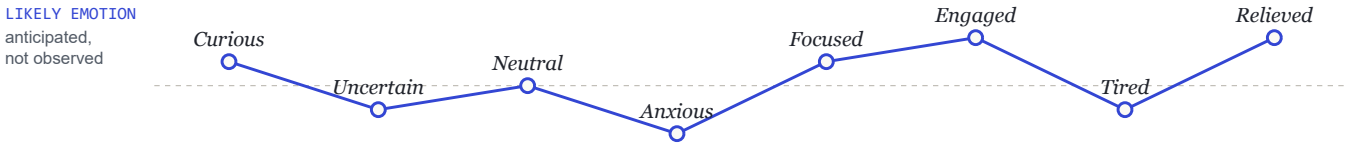
	PC · INSTRUCTOR — RUNNING A TEACHING SESSION	PD · TECHNICIAN — HANDLING A FAULTY CAMERA KIT
Likely says	Has everyone checked in? Which station is free?	When did this stop working? Who had it last?
Likely thinks	Two kits are probably incomplete again.	Who else booked this kit this week?
Likely does	Walks the room; marks attendance; fixes setups; answers repeated questions.	Tests the device; searches for accessories; updates a log; tells faculty.
Likely feels	Stretched across stations; frustrated by preventable delays.	Interrupted; responsible for problems they could not see coming.
Pains	No readiness signal; manual attendance; inconsistent evidence quality.	Symptom-free reports; unknown last user; no link from a fault to future bookings.
Gains	Readiness at a glance; attendance from check-in; rubric linked to evidence.	Structured reports with asset ID; automatic impact on bookings; verification record.

FIG. 03 Empathy map for the first-run student during the vision experiment, with the instructor and technician maps beside it. Every entry is an anticipated hypothesis from the Phase 2 record. No participant said, thought or felt these; zero participants were studied.

SECTION 03 · RESEARCH FIGURES

User journey map

One student, one vision experiment, eight stages: where the anticipated low points are and what the design offers at each.



	01 Discover	02 Prepare	03 Reserve	04 Arrive & check in	05 Capture	06 Analyse	07 Document	08 Submit & return
ACTIONS	Opens assigned experiment	Reads objective, checks prerequisites	Picks time, reserves full kit	Finds station CV-03, verifies kit	Sets lighting and camera, captures	Runs detection, compares lighting conditions	Records observations and limits	Submits revision, returns kit
TOUCHPOINTS	P02, P04	P05	P06, P07	Signage, P10, P11	Capture zone C1, P12	P14	P15	P16, P11 return
PAIN POINTS	Too many similar experiments	Kit list unclear (H01)	Room booked, devices not (H02)	Missing accessory (H01)	Light changes mid-capture (H10)	Confidence read as correctness (H12)	Settings lost (H03)	Unclear if submission is final
OPPORTUNITIES	Topic filters, duration, prerequisites	Checklist that transfers to booking	Complete-kit reservation with conflicts	Same manifest at check-in; report issue inline	Recording indicator; settings captured automatically	Confidence vs ground truth explained	Evidence embeds with provenance	Stable revision + return checklist

FIG. 04 Student journey for vision experiment E06 (J1). Emotions and pain points are anticipated hypotheses (H-numbered in the Phase 2 record), not observations. Touchpoint codes refer to the page and component specifications.

Workspace evidence: research/U02_RESEARCH_SYNTHESIS.md; research/PHASE_02_RESEARCH_AND_EXPERIENCE.md; submission/source/report-figures.json

SECTION 04

4. UI/UX Methodology

The project translates source evidence into requirements, compares alternatives, builds annotated wireframes and then implements the selected local workflow. The original phase-01 through phase-10 work and the later U01 through U10 upgrades are distinct histories. Dated delivery records retain the chronology instead of rewriting early plans as completed studies.

U01 preserves identities, fixtures, geometry and original entrypoints. U02 revises research under the desk-only scope. U03 migrates local functional parity to React. U04 adds the dashboard, observation provenance and sustainability ledger. U05 adds scientific rehearsals and fault consequences. U06 grounds local AI. U07 tests an independent operations desk. U08 completes authored art and physical proposals. U09 checks and refines the current implementation. U10 assembles this submission and its demonstration handoff.

Review combines designer walkthroughs, source integrity, domain tests, browser evidence and rendered artifact checks. The fourteen system scenarios examine conflicts, retries, retained context, policy changes, fault isolation, interrupted verification, unavailable models, public projections and scene lifecycle. A scenario's local closure is separate from a connected service or a human outcome. Changes receive preserved before evidence, bounded implementation, retest evidence and a stated residual limit.

REVIEW METHOD	WHAT IT ESTABLISHES	WHAT IT CANNOT ESTABLISH
Source and unit checks	Rules, scientific identities and immutability	Institutional authorization
Browser/designer walkthrough	Rendered actions, messages and fallback paths	Student comprehension
Automated accessibility scan	Covered machine-detectable conditions	Full WCAG or spoken AT usability
Rendered report/deck review	Pagination, legibility and layout	A human presenter's delivery pace

Workspace evidence: planning/U02_DELIVERY_RECORD.md; planning/REMAINING_WORK_AUDIT_2026-10-04.md; planning/U09_DELIVERY_RECORD.md

SECTION 05

5. Design Process

The visual system uses editorial typography, restrained blue accents, paper-toned surfaces and explicit task hierarchy. Forty component families and eighty-seven Figma variants support consistent state language. Student work starts with a named experiment. Staff tools use separate destinations. Task headings, source labels, recovery messages and next actions carry the information hierarchy across desktop, narrow and public views.

The physical and digital artifacts share canonical identities. C1-CAM and both C1 lights appear in the kit, map, fault impact and standby review. P4-D supplies the model coordinates and plan layers. REF-2HZ-v1 supplies the signal calculations and art. E06 lighting images remain authored illustrations. This shared source avoids inconsistent diagrams without claiming that a model measures the physical installation.

The final audit found three bounded UI issues. The typed-ID field omitted its explicit text type and missed the intended 16 px/44 px styling. Product copy still named older upgrade stages. The QA scan retained another route's result after navigation. The refinement corrects all three, preserves the previous release and adds a visible exact-text export alternative. Copied scene modules now own frame cancellation and resource disposal. Original models and scientific calculations remain preserved.

Physical proposals include six wayfinding/safety sign studies, five canonical plan layers and specifications for maintainable lighting, ventilation, protected power, repairable furniture and traceable materials. Trade-offs and responsible roles accompany each proposal. Procurement, circuit mapping, occupied-room measurement and qualified review remain outside this concept delivery.

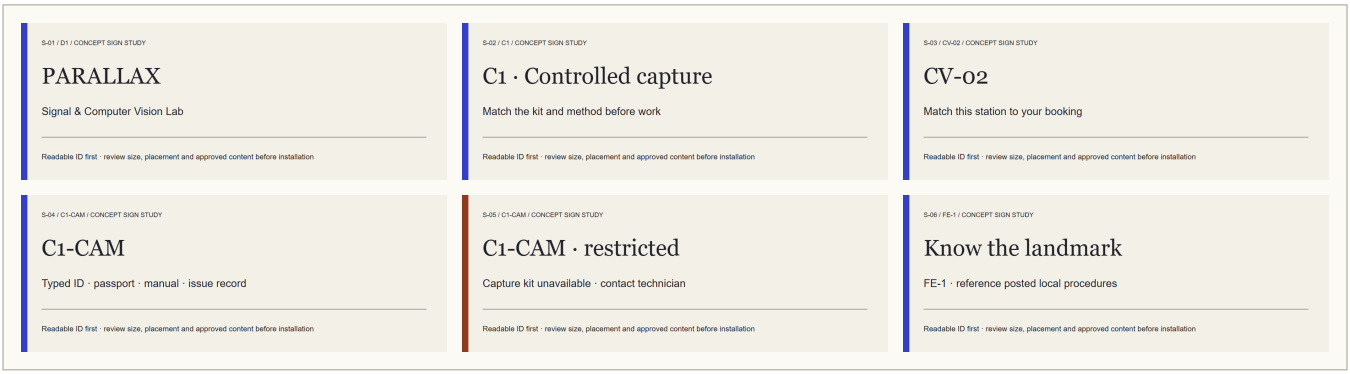


FIG. 05 Six wayfinding and safety sign studies, S-01 to S-06. Conceptual designs; no signage has been installed or approved.

Workspace evidence: design/system/components.json; planning/U08_DELIVERY_RECORD.md; apps/lab-web/evidence/u09-final/mobile-after.json; planning/revisions/U09-before-implementation/manifest.json

SECTION 06

6. Wireframes

The fifteen annotated boards cover preparation, booking, arrival, evidence comparison, notebook, submission, technician recovery, public display and supplementary faculty/asset tasks. Twelve original block layouts remain preserved. Three supplementary boards cover cohort readiness, attendance correction and asset movement. An annotation names the relevant requirement, source, decision and implementation boundary.

WF03 shows complete-kit booking and conflict recovery. Its student-facing refusal identifies the resource clash and retains a compatible alternative. WF06 makes retained evidence, changed conditions, uncertainty and student reasoning visible. WF11 separates report, classification, restriction, repair, verification and restoration. The final app tests these local states, while the structural boards do not imply every hypothetical faculty interaction has a connected backend.

The editable board data and generator remain in design/wireframes. The local review page provides all fifteen SVGs and their source annotations. The Figma file retains the high-fidelity library and linked prototype paths. The current submission index points to these artifacts and the React edition without replacing the historical designs.



FIG. 06 WF03: authored booking structure. Source annotations remain in the complete local wireframe packet.

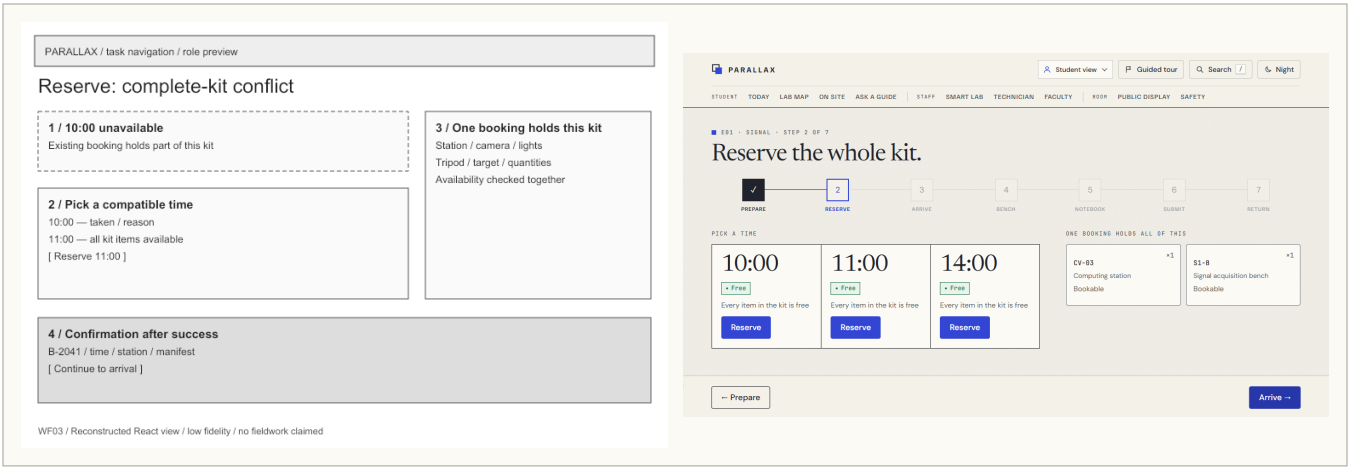


FIG. 07 From wireframe to interface. Left: WF03, the authored booking structure. Right: the reservation step as built, with the same time slots and whole-kit manifest.

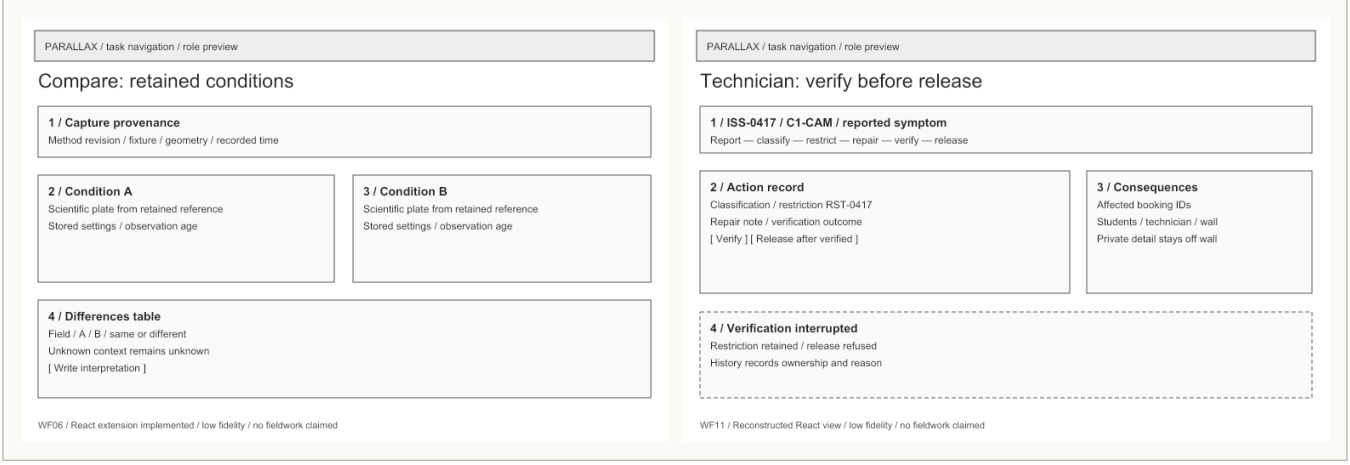


FIG. 08 WF06, retained evidence comparison, and WF11, technician recovery. Authored structures from the fifteen-board packet.

Workspace evidence: design/wireframes/index.html; design/wireframes/boards.json; research/U02_DESK_REVIEW.md; design/figma-build/README.md

SECTION 07

7. Final Prototype

The current React edition delivers student E01/E06 paths, faculty and technician views, booking/check-in, typed equipment lookup, asset/calibration records, the linked map and a public projection. Local browser persistence preserves drafts and supports conflict/recovery behavior. Prepared actors and receipts are examples. No institutional account, authoritative booking database, real attendance service or grade delivery is connected.

The smart dashboard combines the same local resource state with explicit source context. Six observation scenarios distinguish missing, recent, delayed, invalid and disconnected records. The energy ledger separates watts from kilowatt-hours and preserves missing intervals. A standby proposal protects active dependencies and essential services. It shows modelled upper bounds rather than measured savings.

E01 retains a 2 Hz, 1 V sine reference over one second. Sampling at 3 Hz produces four inclusive samples and a signed -1 Hz alias fitting the same values. The 4 Hz zero-phase boundary is ambiguous. At 20 Hz, twenty-one samples are retained. E06 uses ILL-L1-v1 and ILL-L2-v1 images to discuss illumination, without detector accuracy claims. The separate geometry rehearsal holds P4-D fixed while changing horizontal field of view.

The assistant uses installed local Ollama llama3.2:3b to select published guide sections. The server verifies revision and fingerprint before and after inference and returns the reviewed passage wording with its citation. Missing context, provider failure, cancellation and withdrawn guides fail into bounded recovery/manual reading. The independent operations desk and ten-stage fault sandbox preserve the primary student record.

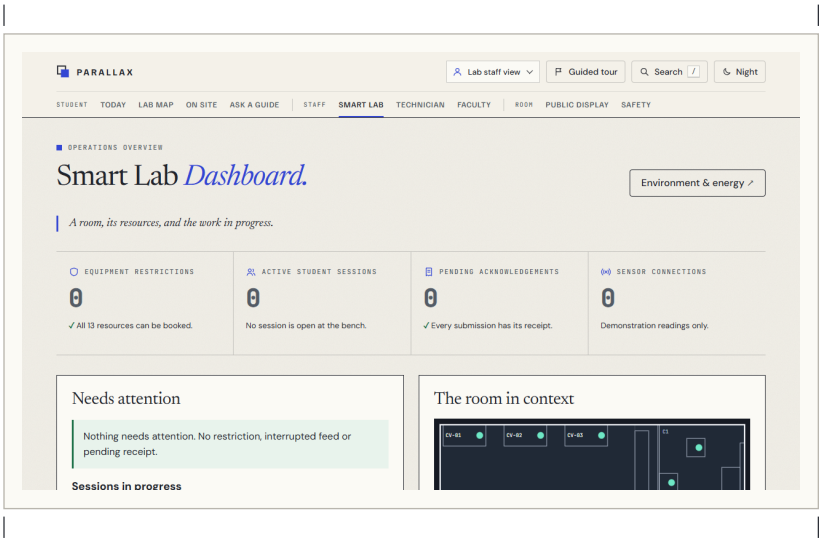


FIG. 09 Current dashboard. Readings are examples; sensor connections are explicitly zero.

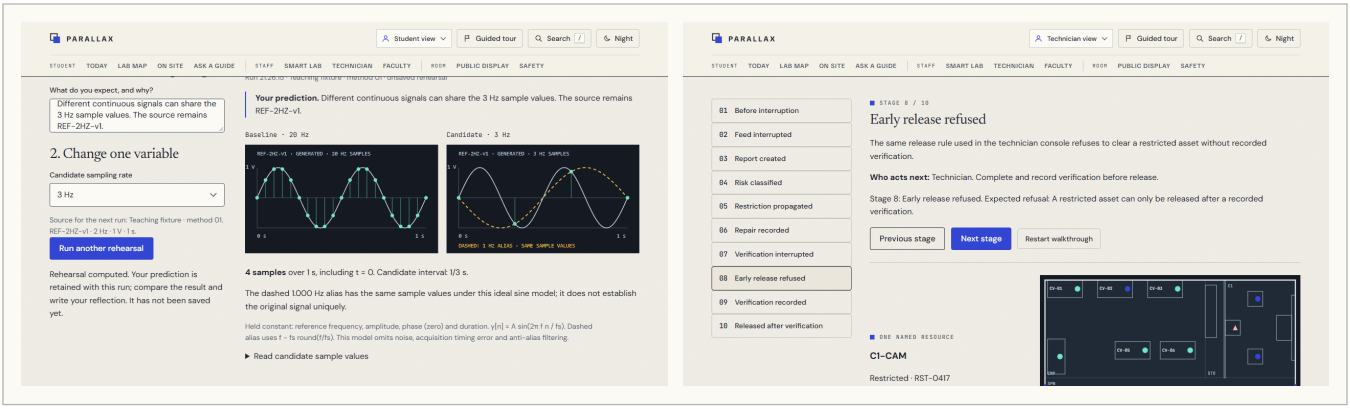


FIG. 10 Left: E01 rehearsal at 3 Hz, a generated teaching fixture. Right: fault sandbox stage 8, where early release is refused.

Smart features, and how real each one is

FEATURE	STATUS	WHAT EXISTS
Digital experiment manual	Working	The question, the kit and a safety check before every session.
Equipment lookup by label	Working	Typing the printed ID opens the record. Camera scanning is a concept.
AI assistant	Working	Local Ollama model answering from reviewed guides, with the cited section.
Digital twin	Working	One geometry drives the live plan and the optional 3D model.
Equipment health	Working	A fault restricts the asset everywhere until a recorded verification.
IoT sensors	Simulated	Example readings with their source and age. No hardware is connected.
Smart energy monitoring	Simulated	Interval energy and a booking-aware standby preview. No meter.
AR / VR learning	Partly	A 3D scene rehearsal runs in the browser. Headset AR is a concept.
Voice assistant	Concept	Proposed for hands-busy bench work. Not built in this prototype.

The same table appears on the prototype dashboard. "Working" means it runs in the local prototype with example records; it does not mean an institutional service is connected.

Workspace evidence: apps/lab-web/PARITY.md; planning/U04_DELIVERY_RECORD.md; planning/U05_DELIVERY_RECORD.md; planning/U06_DELIVERY_RECORD.md; planning/U07_DELIVERY_RECORD.md

SECTION 08

8. Floor Plan

P4-D partitions the assumed 70 m² room into computing 24.96 m², storage 3.12 m², capture 10.92 m², circulation 15.00 m², faculty 3.20 m², collaboration 7.04 m² and signal work 5.76 m². Their sum is 70 m². The source declares metres, an origin at the north-west inner wall, x east and y south. The design retains a 1.5 m spine and a 1.2 × 0.9 m accessible workstation approach.

The C1 camera-to-stage distance is 2.05 m under the planning model. Entry/exit, first aid, extinguisher, electrical isolation, CCTV and ventilation appear as labelled conceptual landmarks. Power/network and installation layers identify proposed service locations. The window W1 and the collaboration zone inform daylight and ventilation questions while the capture area requires repeatable illumination.

Geometric checks confirm furniture stays inside the envelope, does not overlap the spine or each other, and retains the declared approach space. These checks establish model fit. They do not certify installed clearances, escape capacity, electrical protection, CCTV governance, airflow or safe occupancy. Whole-class teaching should use a reviewed rotation plan until actual capacity and supervision are established. The final handoff includes the editable Blender model, GLB views, renders and canonical geometry.

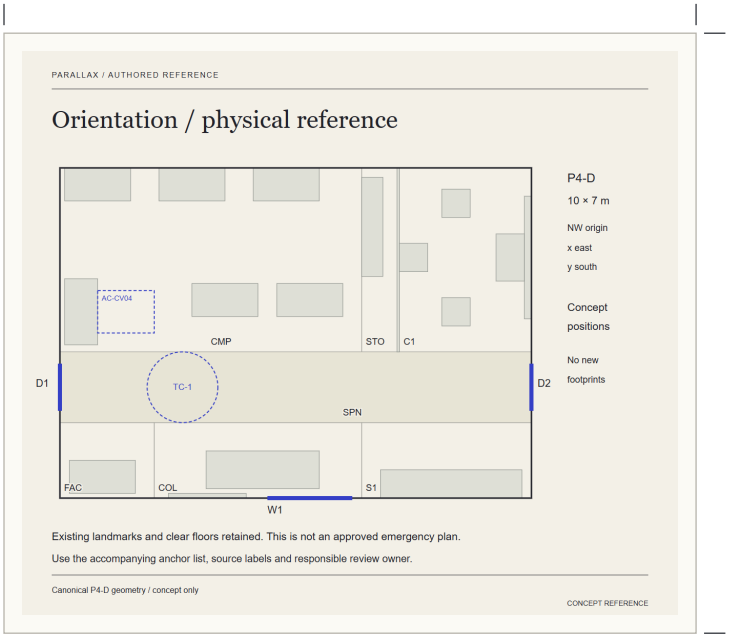


FIG. 11 Canonical P4-D orientation plan. Dimensions and landmarks describe an assumed design envelope.

Workspace evidence: spatial/lab_geometry.json; spatial/exports/upgrade-validation.json; planning/U04_SUSTAINABILITY_SPECIFICATION.md; apps/lab-web/public/studio/reference.md

SECTION 09

9. Technology Stack

The current client uses React 19.3.0, TypeScript 7.0.2, Vite 8.3.2 and React Router 8.4.0 with pinned package-lock dependencies. Typed domain adapters preserve legacy fixture/rule behavior while cloning mutable input. Local storage uses versioned keys for the primary React example, rehearsals and operations exercise. These stores are local records and cannot implement shared multi-client transactions.

Three.js and retained P4-D geometry power optional scene views. Accessible 2D/list/text alternatives accompany them. The refined generated copies own animation frames, resize observation, controls and disposable GPU resources. The original scene/model source is preserved. QA-only route/scene/performance receipts remain explicit and collect no external telemetry.

The Node assistant service binds to loopback port 4192. The edition uses 4180, React development uses 4190 and isolated validation uses 4191. Ollama runs locally at 11434 with the installed completion model. The adapter rejects remote hosts/models and browser evidence. Reviewed guide JSON supplies facts and lifecycle metadata. The service has anonymous local session isolation, bounded concurrency and in-memory request recovery, rather than institutional authentication.

The handoff distinguishes a prebuilt assessor preview from a source rebuild. Node/npm are needed for serving or rebuilding. An Internet connection is needed only to restore missing development dependencies. Ollama and the installed model are optional for live hints because the manual remains readable. No API key or cloud provider is required. Rebuilding or replacing the model requires the documented regression/evaluation checks.

LAYER	ACTUAL RESPONSIBILITY
Fixtures and geometry	Canonical IDs, source reference and assumed room
Domain / local persistence	Immutable records, local rules, drafts and retry state
React / Three.js	Task views, declared models and readable alternatives
Node / Ollama	Public-guide retrieval, passage selection and source validation
Figma / Blender / documents	Editable design, physical model and assessor evidence

Workspace evidence: apps/lab-web/package.json; apps/lab-web/package-lock.json; apps/lab-server/README.md; submission/README.md

SECTION 10

10. Conclusion

PARALLAX now supplies a coherent local proposal covering the assignment's physical and digital families, with a preserved original baseline, current prototype, scientific/source boundaries, sustainability specifications and recovery evidence. Its strongest supported outcome is inspectability: a reviewer can follow a source through a requirement, design decision, screen state and technical check.

The final source/build gate passes 139 React tests and 16 assistant service tests. Original checks retain 65 prototype, 25 scene and 16 spatial passes. The fresh narrow typed-ID control measures 16 px and 44 px with visible keyboard focus and no page overflow. Eight same-document scene remounts retain 22 geometries and three textures per renderer. Thirty-six two-frame feedback samples range below 26 ms on the local browser. These observations describe the tested machine and paths, without proving browser-wide leak freedom or general performance.

The fourteen scenario ledger explicitly distinguishes local test support from conditional integration. Late/out-of-order live stream ingestion remains unimplemented because the app has no live stream. This is a documented boundary rather than a fabricated passing sequence. The designer comparison walkthrough checks visible difference, uncertainty and next action without claiming student comprehension.

The remaining external verification list names spoken screen-reader use, actual browser zoom, a second browser/physical device, native download completion and a human presenter's rehearsal. Tool availability does not turn them into passes. Connected hardware, authenticated shared operations, institutional policies and physical installation approval remain future integrations. Team identities, actual contributions and the deadline year must be supplied by the team before institutional submission. No participant work or certification is invented.

Workspace evidence: planning/U09_DELIVERY_RECORD.md; submission/EVIDENCE_INDEX.md; submission/EXTERNAL_CHECKS.md; apps/lab-web/evidence/u09-final/performance.json; apps/lab-web/evidence/u09-final/scene-lifecycle.json

APPENDIX A

References and reading scope

The U02-R2 source register records the actual reading depth and limitations. Access dates are kept in the source register. Full text availability varies. These sources support the design rationale, without validating local user or site outcomes.

S01 - N. G. Holmes, C. E. Wieman and D. A. Bonn (2015). Teaching critical thinking. Reading scope: full.

https://arxiv.org/abs/1508.04870

S02 - B. M. Zwickl, T. Hirokawa, N. Finkelstein and H. J. Lewandowski (2014; preprint revised 2 May 2014). An epistemology and expectations survey about experimental physics: Development and initial results. Reading scope: abstract.

https://arxiv.org/abs/1310.1055

S03 - Martin M. Stein, Emily M. Smith and N. G. Holmes (2018 preprint). Confirming what we know: Understanding questionable research practices in intro physics labs. Reading scope: full.

https://arxiv.org/abs/1807.04716

S04 - C. M. Rathnayaka et al. (full author list not verified) (Online 2023; journal issue 2024). Preparative pre-laboratory online resources for effectively managing cognitive load of engineering students. Reading scope: indexed-partial.

https://www.tandfonline.com/doi/abs/10.1080/03043797.2023.2204813

S05 - Max Schröder, Susanne Staehlke, Paul Groth, J. Barbara Nebe, Sascha Spors and Frank Krüger (31 January 2022). Structure-based knowledge acquisition from electronic lab notebooks for research data provenance documentation. Reading scope: abstract.

https://pubmed.ncbi.nlm.nih.gov/35101121/

S06 - University of Edinburgh · Confocal and Advanced Light Microscopy (Undated live page). CALM user guidelines and rules. Reading scope: full.

https://clinical-sciences.ed.ac.uk/qmri-facilities/calm-confocal-and-advanced-light-microscopy/user-guidelines

S07 - University of Edinburgh · School of Biological Sciences (17 June 2024). Light Microscopy Core: Make a booking. Reading scope: full.

https://biology.ed.ac.uk/drp/light-microscopy-core/make-a-booking

S08 - W3C (Recommendation; current linked edition). Web Content Accessibility Guidelines (WCAG) 2.2. Reading scope: full.

https://www.w3.org/TR/WCAG22/

S09 - UK Health and Safety Executive (Live guidance). Using CO2 monitors. Reading scope: full.

https://www.hse.gov.uk/ventilation/using-co2-monitors.htm

S10 - University College London (Live programme page). LEAF — Laboratory Efficiency Assessment Framework. Reading scope: full.

https://www.ucl.ac.uk/about/sustainability/campaigns-and-initiatives/leaf-laboratory-efficiency-assessment-framework

S11 - Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı and Rei Mariman (2025; correction notice 20 August 2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. Reading scope: full-main-correction-unread.

https://www.pnas.org/doi/10.1073/pnas.2422633122

S12 - Greg Kestin, Kelly Miller, Anna Klales, Timothy Milbourne and Gregorio Ponti (3 June 2025). AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting. Reading scope: full.

https://www.nature.com/articles/s41598-025-97652-6

S13 - Anonymous public contributors, r/labrats (Public thread; displayed relative dates not converted to an exact publication date). Shared lab equipment booking — Am I alone?. Reading scope: full.

https://www.reddit.com/r/labrats/comments/1pn49rd/shared_lab_equipment_booking_am_i_alone/

S14 - Anonymous public contributors, r/labrats (Public thread; displayed relative dates not converted to an exact publication date). Scheduling Shared Equipment. Reading scope: full.

https://www.reddit.com/r/labrats/comments/17cgsjp/scheduling_shared_equipment/

APPENDIX B

Requirement and rubric evidence

The original U02-R2 traceability contains all twenty requirements. This appendix preserves its design-stage boundaries; current implementation coverage is indexed in submission/EVIDENCE_INDEX.md and apps/lab-web/PARITY.md. Participant collection is excluded by the agreed desk-research scope; local user outcomes remain unmeasured.

ID	REQUIREMENT	BOARD	U02-R2 BOUNDARY
R01	Versioned kit with accessory quantities	WF02;WF03;WF04	Local kit demonstrator; real quantities/policy unverified
R02	Reserve all required resources for the same interval	WF03	Local complete-kit conflict model; shared transaction and local policy open
R03	Preserve conflict choices and offer alternatives	WF03	Local conflict/recovery demonstrated; local user outcomes unmeasured
R04	Check-in reuses manifest and records attendance	WF04	Local arrival demonstrated; institutional attendance integration open
R05	Capture stores source, settings and timestamp automatically	WF05;WF06	Immutable example context implemented; live device acquisition absent
R06	Processing output links input, parameters and version	WF05;WF06;WF07	Demonstrator source/parameters retained; full processing parity open
R07	Confidence labelled separately from correctness	WF05;WF07	Illustrative evidence boundary exists; local comprehension untested
R08	Notebook evidence identity and provenance	WF07	Local notebook/evidence route; full embed parity open
R09	Submission freezes stable manifest revision	WF08	Local frozen submission/receipt implemented; server authority absent
R10	Issue preselects asset and captures symptoms	WF05;WF11	Local seeded issue and technician route; broader reporting parity open
R11	Restriction updates passport, booking and map together	WF03;WF09;WF11; WF12	Local restriction projections; authoritative cross-user synchronisation open
R12	Notify affected bookings with an alternative	WF03;WF11	Design/legacy demonstration; real notification delivery absent
R13	Restore availability only after recorded verification	WF11	Local verification gate demonstrated; authorised physical release unverified

ID	REQUIREMENT	BOARD	U02-R2 BOUNDARY
R14	Instructor cohort readiness before session	WF13	Original faculty design plus dedicated low-fidelity reconstruction; React flow/local validation open
R15	Attendance correction actor, reason and time	WF14	Original attendance design plus dedicated low-fidelity reconstruction; policy/React flow open
R16	Asset move from/to, actor, time and reason	WF15	Original operational requirement plus dedicated movement wireframe; React flow and real inventory open
R17	QR and manual ID resolve to same record	WF04;WF12	Typed path demonstrated; actual QR scanner/device validation open
R18	Shared zone/asset IDs across signage, plan and software	WF09;WF12	Canonical assumed geometry reused; installed signage/site verification absent
R19	Every 3D task has list or 2D equivalent	WF12	Existing map fallback; complete all-task/access review open
R20	Capture zone avoids through-circulation	WF12	Assumed 10 × 7 m planning concept; not a surveyed or approved physical layout

APPENDIX C

Assessment and handoff

CRITERION	MARKS	EVIDENCE
User Research	10	U02 register, synthesis, personas, empathy map and journey map (section 3 figures)
Problem Analysis	10	Twenty requirements and eight decisions
Space Planning	15	P4-D geometry, model, clearances and plan layers
UI Design	15	Fifteen boards, Figma library and current task views
UX Design	15	Fourteen scenario ledger, recovery and retained-context paths
Innovation & Smart Features	10	U04 provenance/energy, U05 models, U06 real local AI
Accessibility & Safety	10	Typed/list/manual paths, scans, keyboard evidence, sign studies and qualified limits
Prototype & Demonstration	10	Source/build checks, walkthrough and static backup
Presentation & Documentation	5	This report, editable deck, notes and current index

Contribution and administrative declaration

The workspace contains design, code, generated assets and verification records. File authorship and commits cannot establish individual team contributions here because this checkout has no Git history and no confirmed team list. The submission does not assign work to unnamed students. The team should attach its own truthful declaration before institutional submission.

The assignment states a team of three students and a 5 August deadline, without a year. No date or deadline status is inferred. The local launcher and README provide the clean start and manual fallback. External device, assistive-technology and site checks remain explicitly listed in EXTERNAL_CHECKS.md.